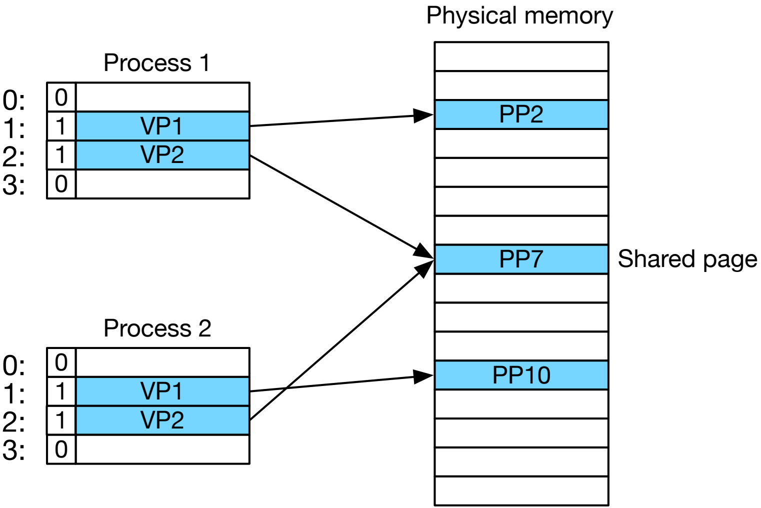
Lecture 23: Virtual Memory II

Brennon Brimhall

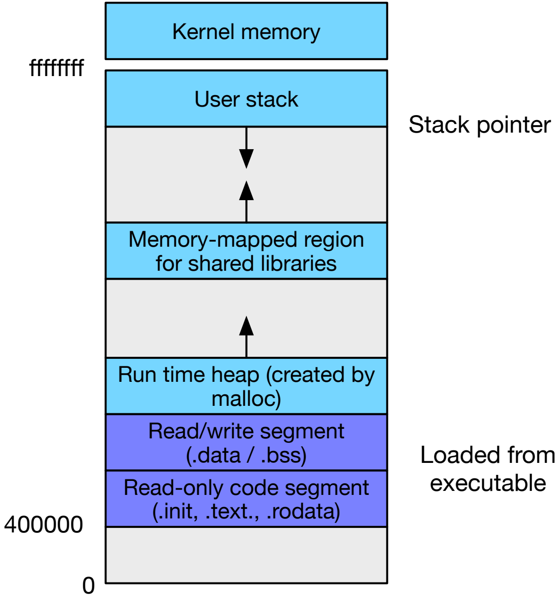
7 July 2026

Memory management

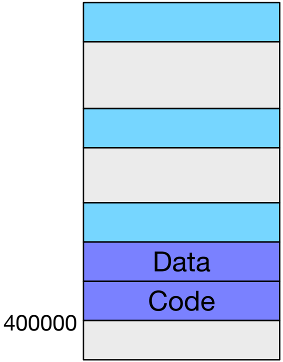
One Page Table per Process



Process Address Space



Simplified Linking



- Each process has its code in address 0x400000
- Easy linking: Linker can establish fixed addresses

Simplified Loading

- When loading process into memory...
- Enter .data and .text section into page table

Simplified Loading

- When loading process into memory...
- Enter .data and .text section into page table
- Mark them as invalid (= not actually in RAM)

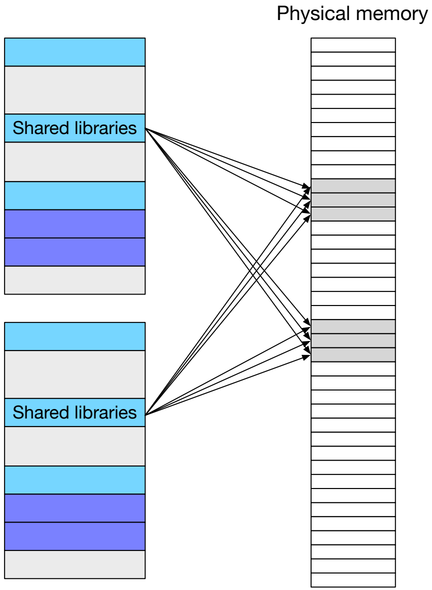
Simplified Loading

- When loading process into memory...
- Enter .data and .text section into page table
- Mark them as invalid (= not actually in RAM)
- Called memory mapping (more on that later)

Simplified Sharing

Shared libraries used by several processes: e.g., stdio providing printf, scanf, open, close, ...

Not copied multiple times into RAM



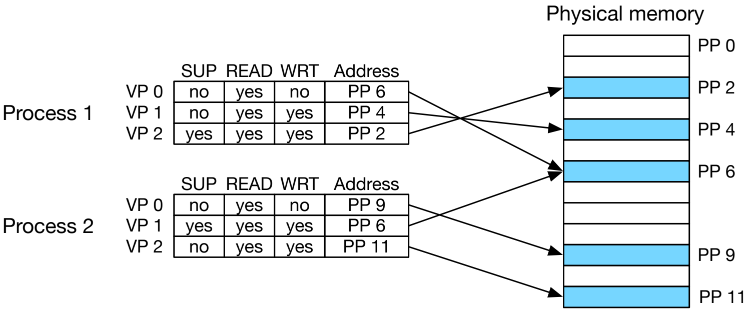
Simplified Memory Allocation

- Process may need more memory (e.g., malloc call)

$\Rightarrow$ New entry in page table

- Mapped to arbitrary pages in physical memory
- Do not have to be contiguous

Memory Protection



- Page may be kernel only: SUP=yes
- Page may be read-only (e.g., code)

Address translation

Address Space

- Virtual memory size: $\mathsf{N} = 2^{\mathtt{n}}$ bytes
- Physical memory size: $\mathsf{M} = 2^{\mathtt{m}}$ bytes
- Page (block of memory): $\mathsf{P} = 2^{\mathtt{p}}$ bytes
- A virtual address can be encoded in n bits

Address Translation

- Task: mapping virtual address to physical address
  - virtual address (VA): used by machine code instructions
  - physical address (PA): location in RAM
- Formally

$$\text{MAP: VA} \rightarrow \text{PA} \cup 0$$

where:

$$\begin{aligned}\text{MAP(A)} &= \text{PA if in RAM} \\ &= \text{0 otherwise}\end{aligned}$$

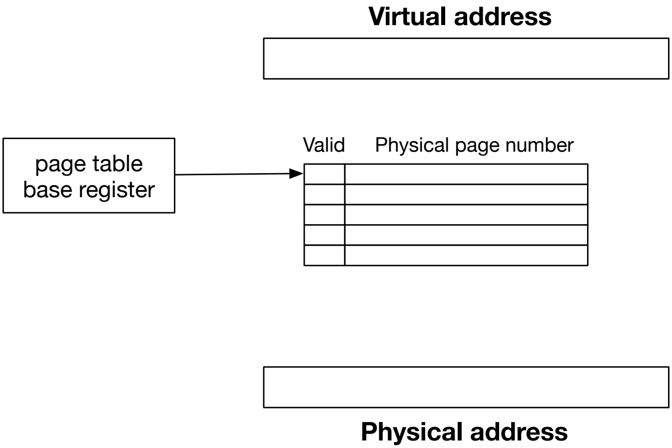
- Note: this happens very frequently in machine code
- We will do this in hardware: Memory Management Unit (MMU)

Basic Architecture

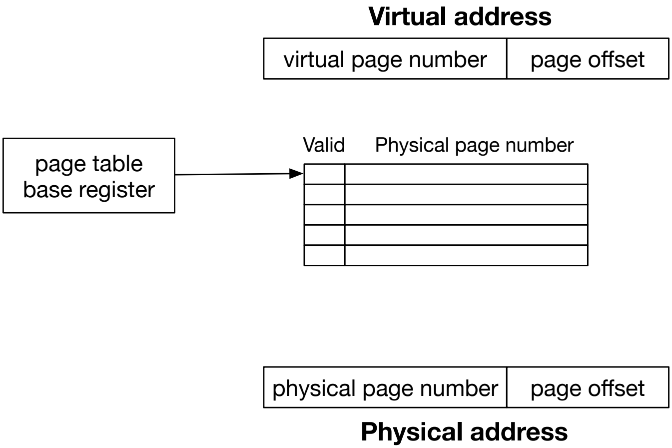
Virtual address

Physical address

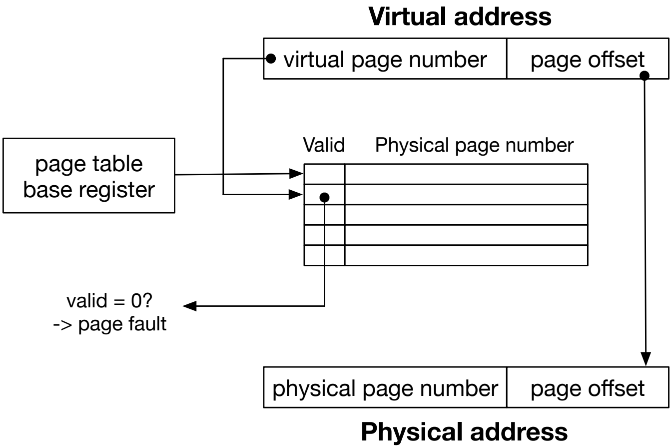
Basic Architecture



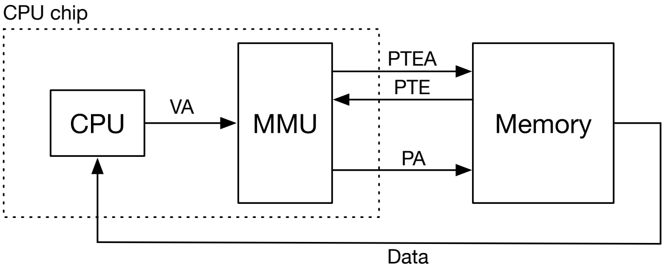
Basic Architecture



Basic Architecture

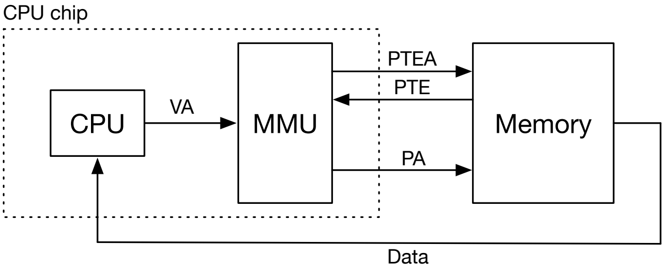


Page Hit



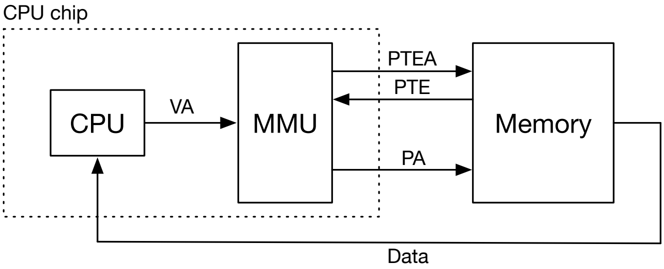
- VA: CPU requests data at virtual address

Page Hit



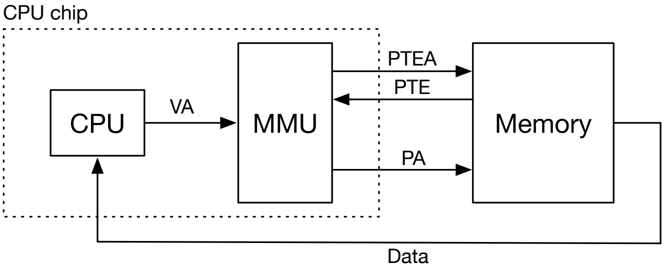
- VA: CPU requests data at virtual address
- PTEA: look up page table entry in page table

Page Hit



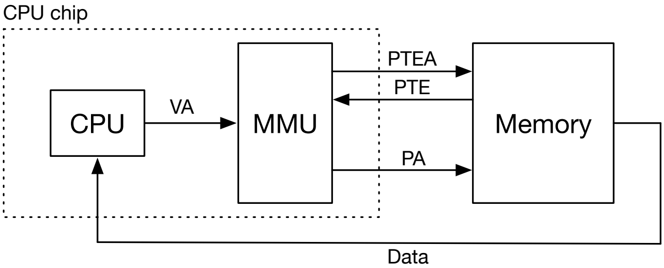
- VA: CPU requests data at virtual address
- PTEA: look up page table entry in page table
- PTE: returns page table entry

Page Hit



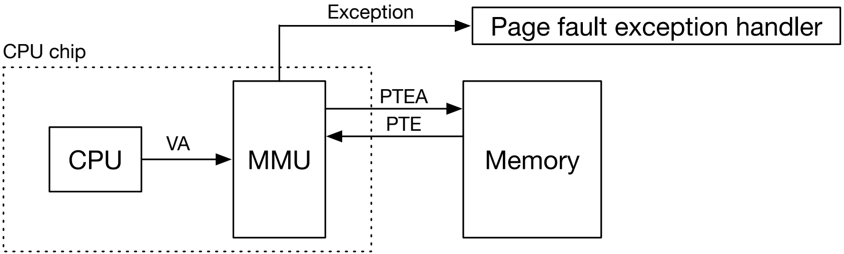
- VA: CPU requests data at virtual address
- PTEA: look up page table entry in page table
- PTE: returns page table entry
- PA: get physical address from entry, look up in memory

Page Hit



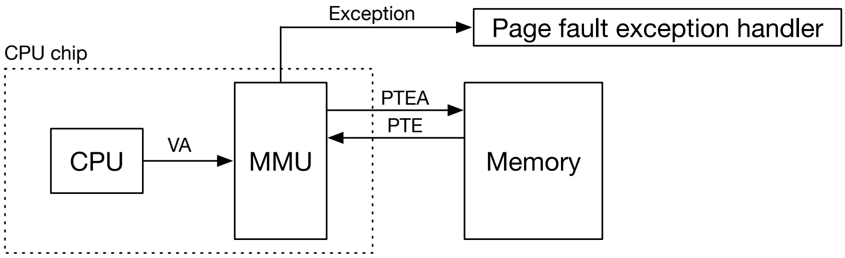
- VA: CPU requests data at virtual address
- PTEA: look up page table entry in page table
- PTE: returns page table entry
- PA: get physical address from entry, look up in memory
- Data: returns data from memory to CPU

Page Fault



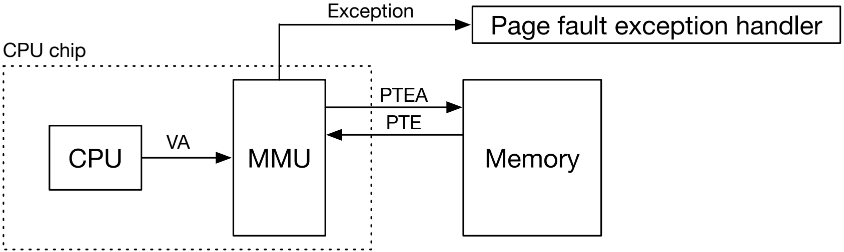
- VA: CPU requests data at virtual address

Page Fault



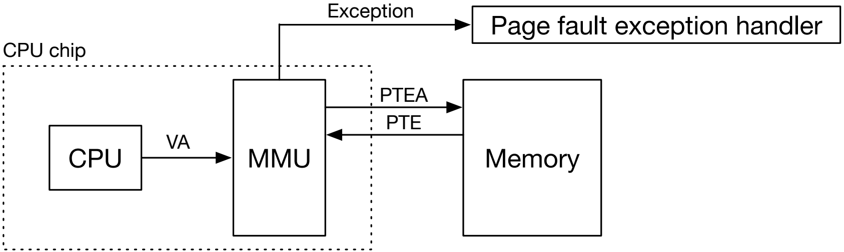
- VA: CPU requests data at virtual address
- PTEA: look up page table entry in page table

Page Fault



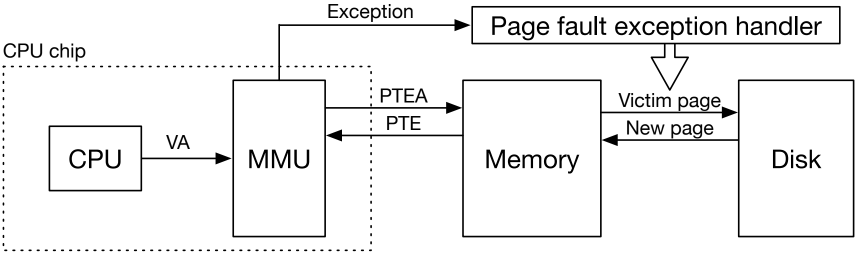
- VA: CPU requests data at virtual address
- PTEA: look up page table entry in page table
- PTE: returns page table entry

Page Fault



- VA: CPU requests data at virtual address
- PTEA: look up page table entry in page table
- PTE: returns page table entry
- Exception: page not in physical memory

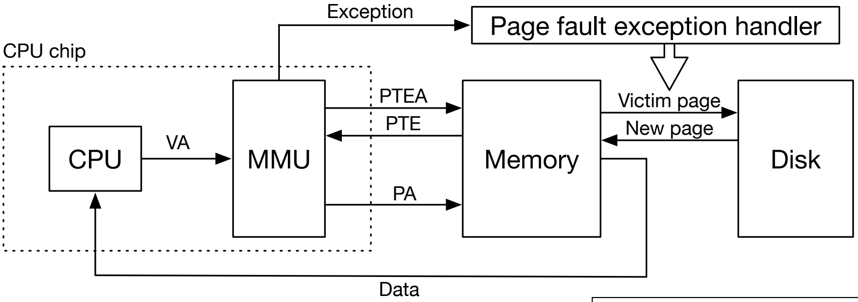
Page Fault



- VA: CPU requests data at virtual address
- PTEA: look up page table entry in page table
- PTE: returns page table entry
- Exception: page not in physical memory
- Page fault exception handler

- victim page to disk
- new page to memory
- update page table entries

Page Fault



- VA: CPU requests data at virtual address
- PTEA: look up page table entry in page table
- PTE: returns page table entry
- Exception: page not in physical memory
- Page fault exception handler

- victim page to disk
- new page to memory
- update page table entries
- Re-do memory request

Page Miss Exception

- Complex task
  - identify which page to remove from RAM (victim page)
  - load page from disk to RAM
  - update page table entry
  - trigger do-over of instruction that caused exception
- Note
  - loading into RAM very slow
  - added complexity of handling in software no big deal

Clicker quiz!

Clicker quiz omitted from public slides

Refinements

Refinements

- On-CPU cache
- Slow look-up time
- Huge address space
- Putting it all together

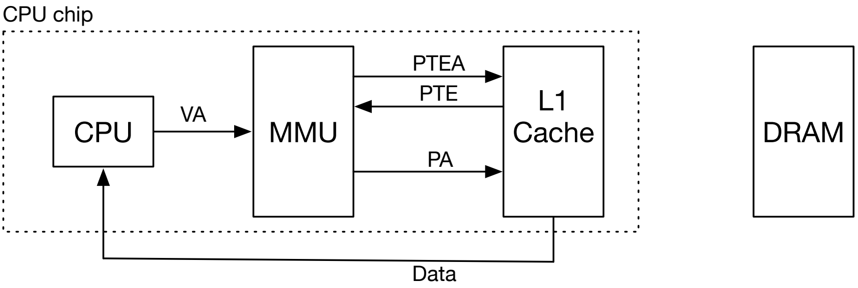
Refinements

- **On-CPU cache**
  $\rightarrow$ **integrate cache and virtual memory**
- Slow look-up time
- Huge address space
- Putting it all together

Integrating Caches and Virtual Memory

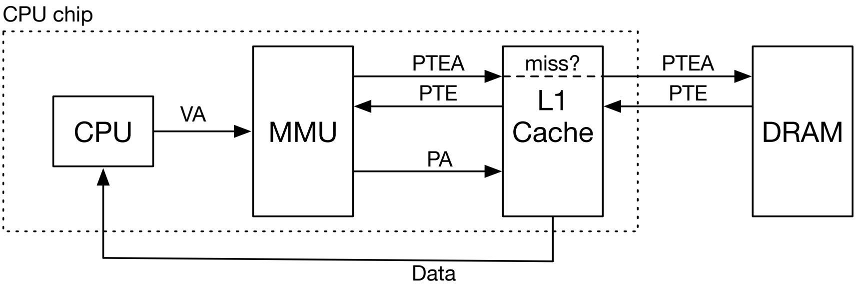
- Note
  - we claim that using on-disk memory is too slow
  - having data in RAM only practical solution
- Recall
  - we previously claimed that using RAM is too slow
  - having data in cache only practical solution
- Both true, so we need to combine

Integrating Caches and Virtual Memory



- MMU resolves virtual address to physical address
- Physical address is checked against cache

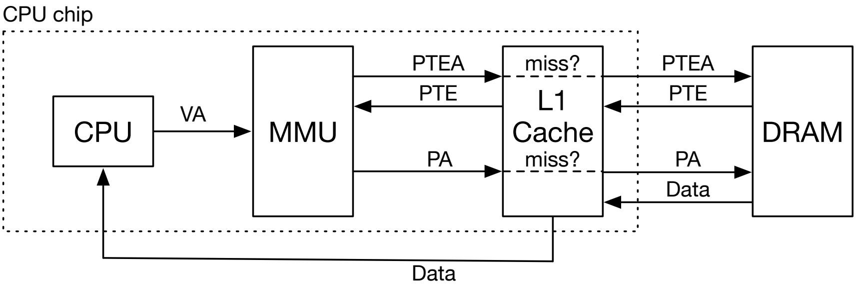
Integrating Caches and Virtual Memory



- Cache miss in page table retrieval?

$\Rightarrow$ Get page table from memory

Integrating Caches and Virtual Memory



- Cache miss in data retrieval?

$\Rightarrow$ Get data from memory

Refinements

- On-CPU cache
  $\rightarrow$ integrate cache and virtual memory
- **Slow look-up time**
  $\rightarrow$ **use translation lookahead buffer (TLB)**
- Huge address space

- Putting it all together

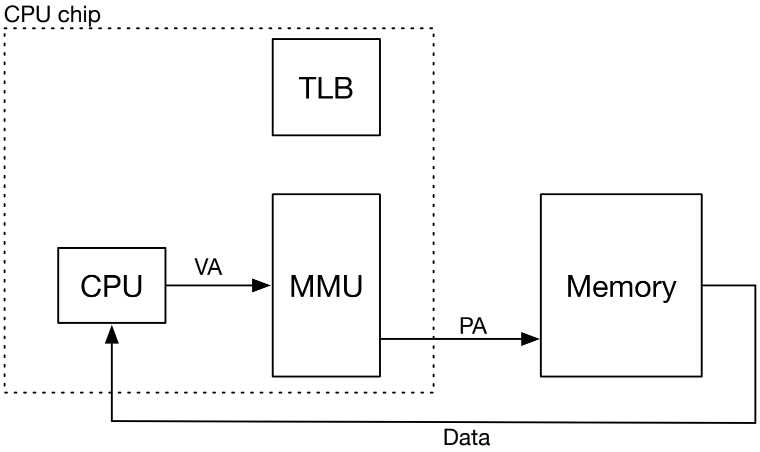
Look-Ups

- Every memory-related instruction must pass through MMU (virtual memory look-up)
- Very frequent, this has to be very fast
- Locality to the rescue
  - subsequent look-ups in same area of memory
  - look-up for a page can be cached

Translation Lookup Buffer

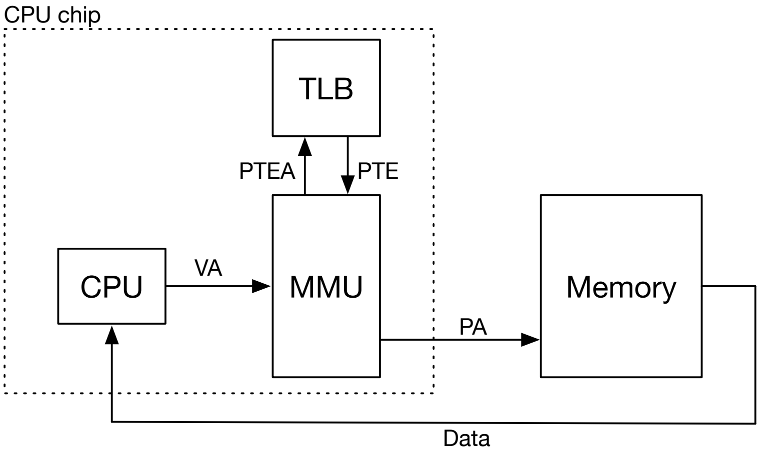
- Same structure as cache
- Break up address into 3 parts
  - lowest bits: offset in page
  - middle bits: index (location) in cache
  - highest bits: tag in cache
- Associative cache: more than one entry per index

Architecture



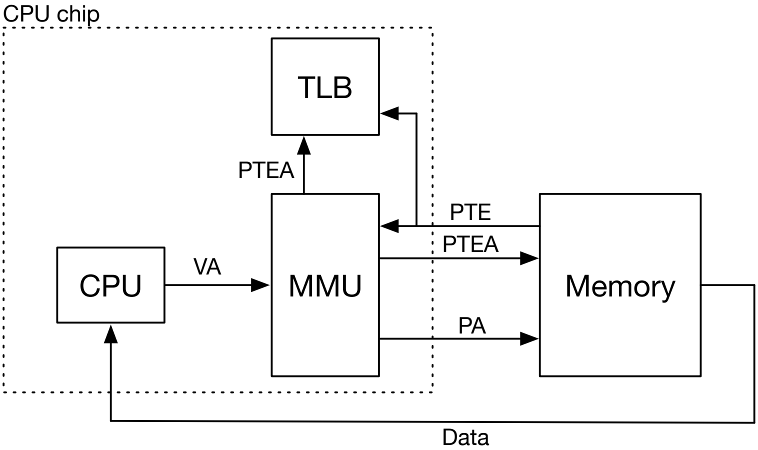
- Translation lookup buffer (TLB) on CPU chip

Translation Lookup Buffer (TLB) Hit



- Look up page table entry in TLB

Translation Lookup Buffer (TLB) Miss



- Page table entry not in TLB
- Retrieve page table entry from RAM

Acknowledgements

Slides adapted from materials provided by David Hovemeyer.